

Kalyani Vemasani

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Artificial Intelligence Machine Learning Engineer

Experienced AI and machine learning engineer skilled in designing and deploying secure Generative AI solutions using Large Language Models and Retrieval-Augmented Generation. Proven expertise in building scalable data pipelines and cloud-native architectures with Python, Spark, and Azure. Strong background in data preprocessing, semantic search, and prompt engineering within the banking domain. Adept at collaborating cross-functionally to deliver compliant, high-performance AI applications that enhance business productivity.

SKILLS

Programming and Scripting: Python, SQL, Bash, HTML, CSS, Pandas, PySpark, NumPy, SciPy.

Cloud Platforms and Services: AWS S3, AWS Lambda, AWS Glue, Amazon Bedrock, Azure Data Factory, Azure Databricks, Azure Data Lake Gen2, Azure Synapse, Azure Functions.

Machine Learning and AI: GPT, Llama, LangChain, LangGraph, LlamaIndex, FAISS, Retrieval-Augmented Generation, Prompt Engineering, Deep Learning, NLP, PyTorch, Scikit-learn, TensorFlow, XGBoost.

Data Engineering and Big Data: Apache Spark, Hadoop, Hive, Kafka, Apache Airflow, Apache NiFi, Snowflake, Azure Event Grid, Docker, Git, CI/CD, MLflow, Linux.

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern, TruLogic Inc, Austin, TX

Jan 2025 - Present | TX, US

- Engineered secure Generative AI applications utilizing Large Language Models and Retrieval-Augmented Generation to optimize knowledge retrieval from extensive banking documentation.
- Constructed and refined RAG pipelines with Python, LangChain, Amazon Bedrock, and FAISS, achieving enhanced semantic search accuracy across diverse document formats.
- Developed robust data preprocessing pipelines using Python, Pandas, and PySpark to clean, normalize, and validate semi-structured and unstructured banking data for embedding generation.
- Enhanced semantic search performance by experimenting with chunking strategies, embedding models, and metadata filtering, improving retrieval relevance and reducing irrelevant results.
- Automated document ingestion, embedding generation, and AI inference workflows using Python scripts and REST APIs, improving operational efficiency and integration with enterprise systems.

Associate Data Engineer, TechBeez

Aug 2020 - Oct 2023 | KA, IN

- Designed and implemented scalable ETL/ELT pipelines with Python, SQL, Azure Data Factory, and Apache Airflow, boosting data processing efficiency for banking datasets.
- Optimized distributed data processing with Apache Spark, PySpark, and Azure Databricks, reducing batch processing times and enhancing scalability for multi-terabyte banking data.
- Created reusable data transformation frameworks that standardized cleaning, normalization, and business rule implementation, accelerating pipeline development and improving data consistency.
- Developed real-time data ingestion pipelines using Apache Kafka, Azure Functions, and Event Grid, enabling near real-time analytics and operational reporting with minimal latency.
- Automated data validation and reconciliation processes, reducing manual verification efforts while ensuring data quality and compliance with banking regulatory standards.
- Constructed RESTful APIs with Python and Flask for secure data access, enforcing authentication and authorization within enterprise banking platforms.
- Enhanced Spark job performance through partition optimization, caching, and query tuning, achieving faster execution times and reduced infrastructure costs.
- Implemented DevOps practices including CI/CD pipelines, Docker containerization, and Git version control, improving deployment consistency and accelerating release cycles.
- Monitored and troubleshooted production data workflows, performing root cause analysis and remediation to maintain data availability and meet strict SLAs in banking operations.

EDUCATION

Masters, University of Maryland, Baltimore County (CGPA: 3.8)

Jan 2024 - Dec 2025 | MD, USA

CERTIFICATIONS

Microsoft Certified Azure AI Engineer Associate.

AWS Certified AI-Practitioner Foundational.